

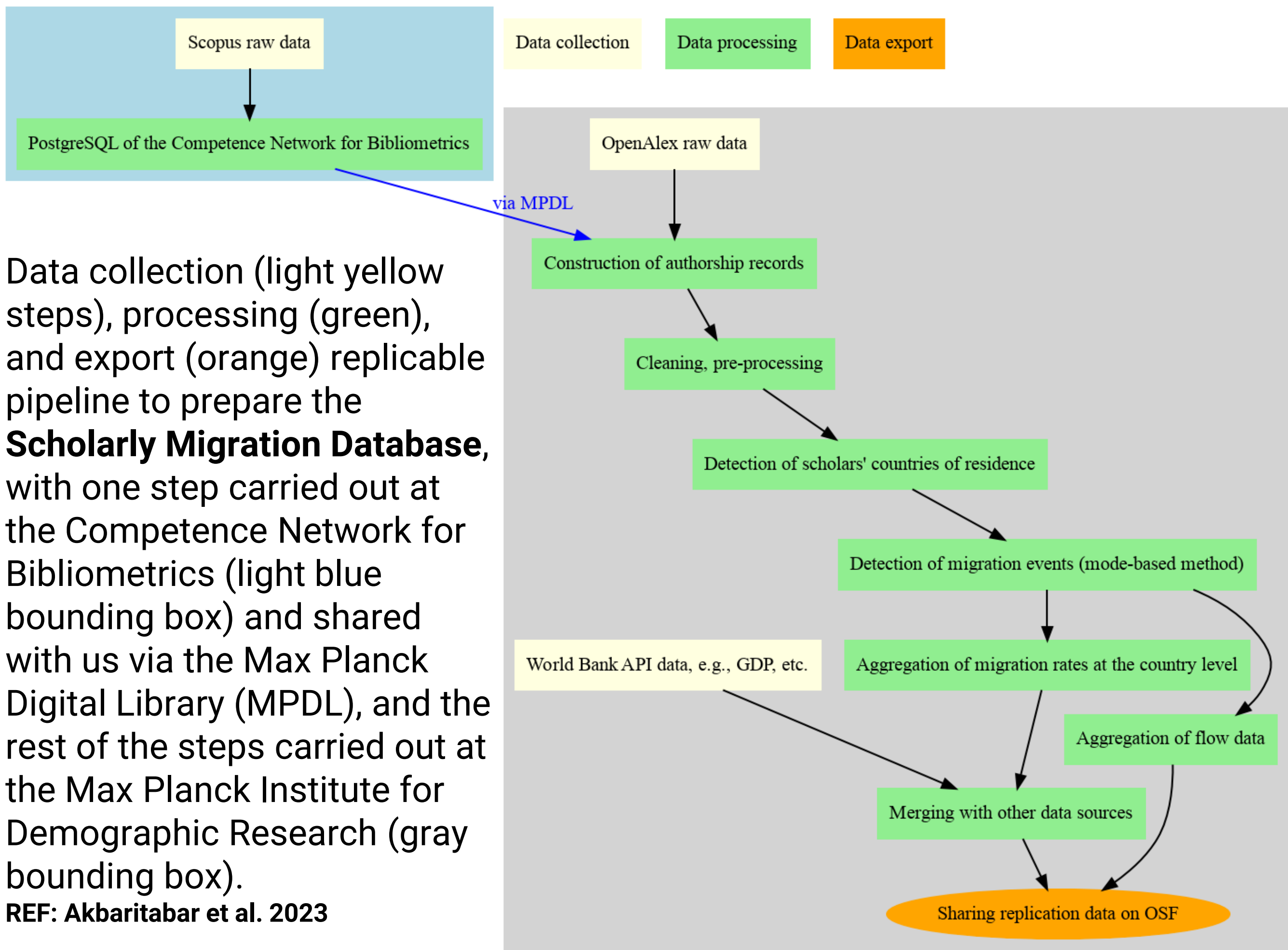


Quantifying the Mobility and Migration of Scholars

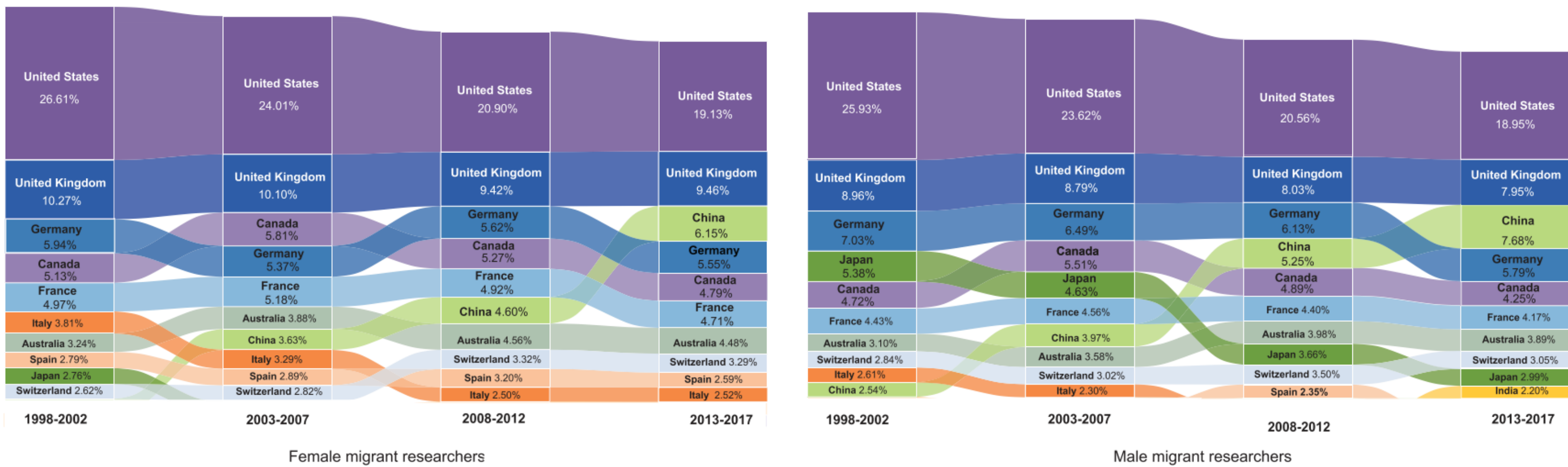
Aliakbar Akbaritabar, the SMD core team, and collaborators

Why? A lack of high-resolution and longitudinal data on migration of scholars worldwide and the unprecedented opportunity to repurpose scientific publications' metadata that spans more than 25 years.

How?

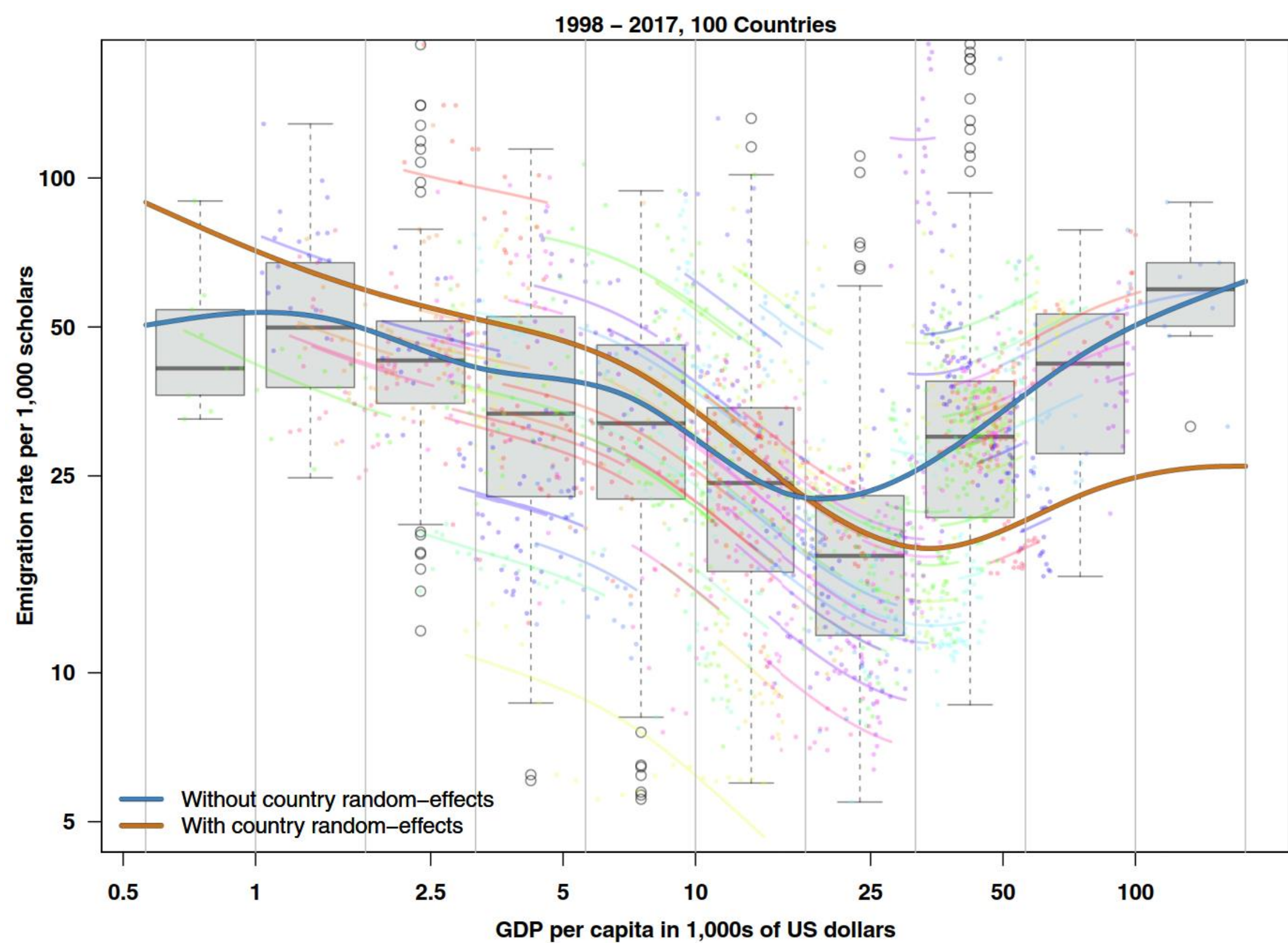


REF: Akbaritabar et al. 2023



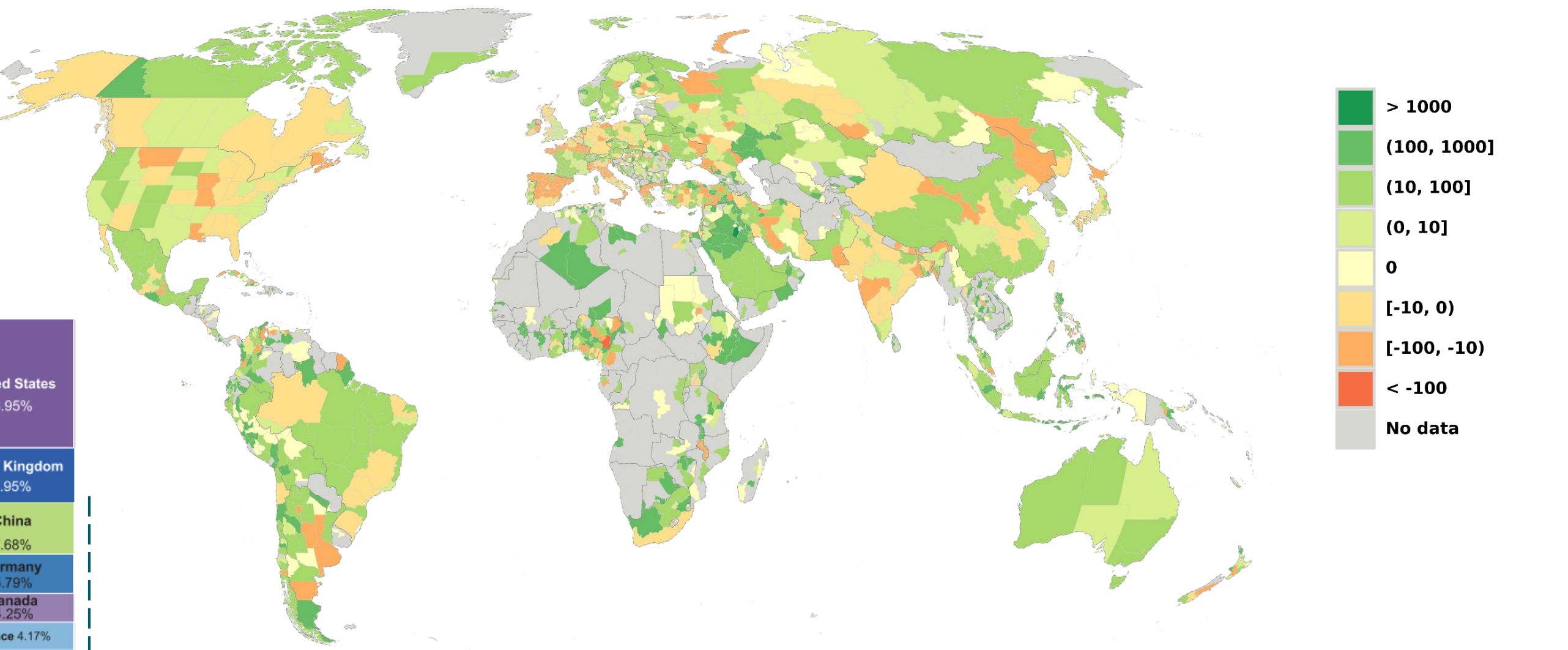
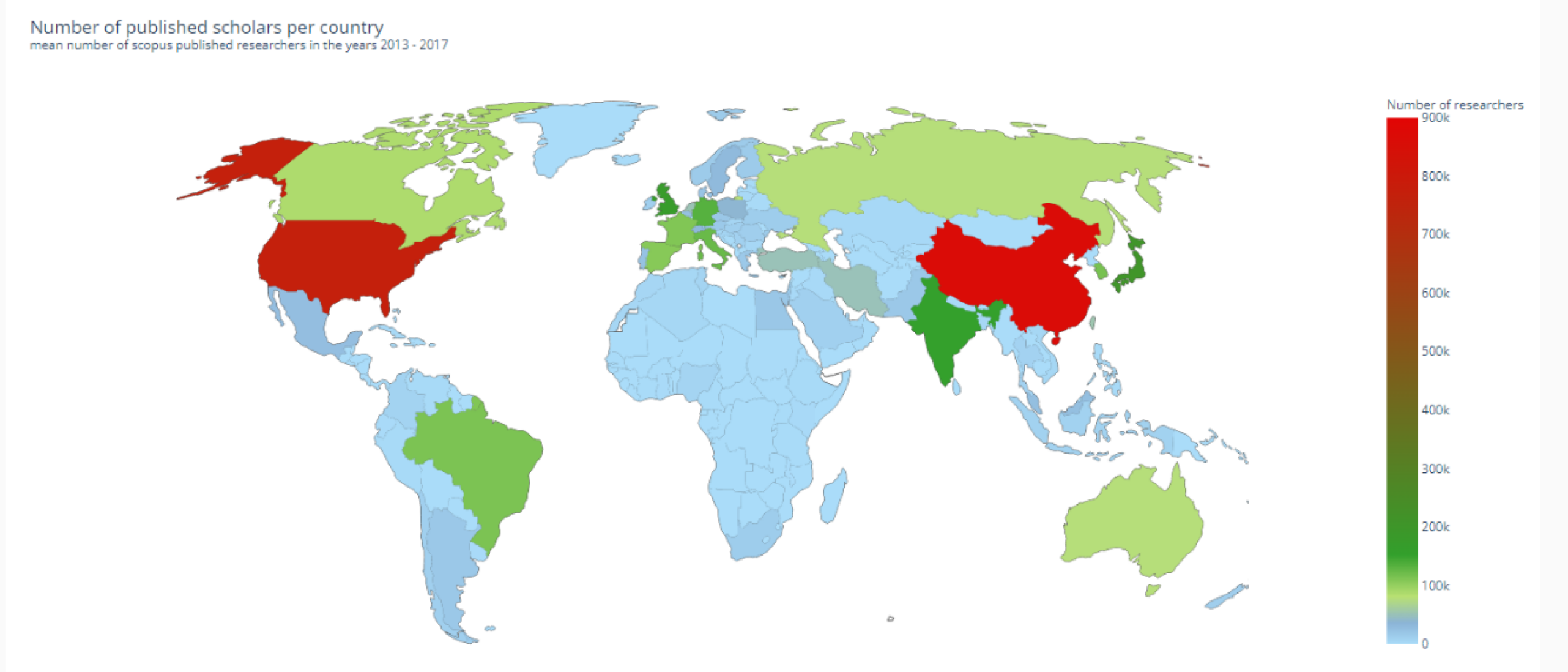
The 10 most preferred destination countries for female mobile researchers (left) and male mobile researchers (right).
REF: Zhao et al. 2023

Scholarly emigration rates (1998 to 2017) as a function of GDP per capita in 2017 constant international dollars at purchasing power parity.
REF: Sanliturk et al. 2023



Scholarly Migration Database Quantifying the Mobility of Scholars

The Scholarly Migration Database provides detailed data about the mobility of scholars. Scientists, journalists and policy analysts who are interested in the movements of published researchers can access high-quality datasets and estimates via this website. The Scholarly Migration Database is prepared and hosted at the Max Planck Institute for Demographic Research in Rostock, Germany.



Combined internal and international net migration rate (NMR) per 1000 scholars at the sub-national level (GeoNames admin-1) worldwide (top) and Europe (right). Colors present the NMR of scholars sent (negative, red colors) or received (positive, green colors) in a year from 2012-2017 as an illustrative example of the 25 years analyzed, and yellow shows a balanced flow.
REF: Akbaritabar et al. 2023

Next?

- The next release of the SMD in 2024 will include disaggregated data by **gender**, **subnational regions** and **OpenAlex** equivalents to Scopus data
- A systematic calculation of return migration
- Considering international scholarly collaboration alongside mobility (i.e., if COVID inspired this as an alternative to geographical mobility?)

More?



Department of Digital and Computational Demography

Laboratory: Migration and Mobility



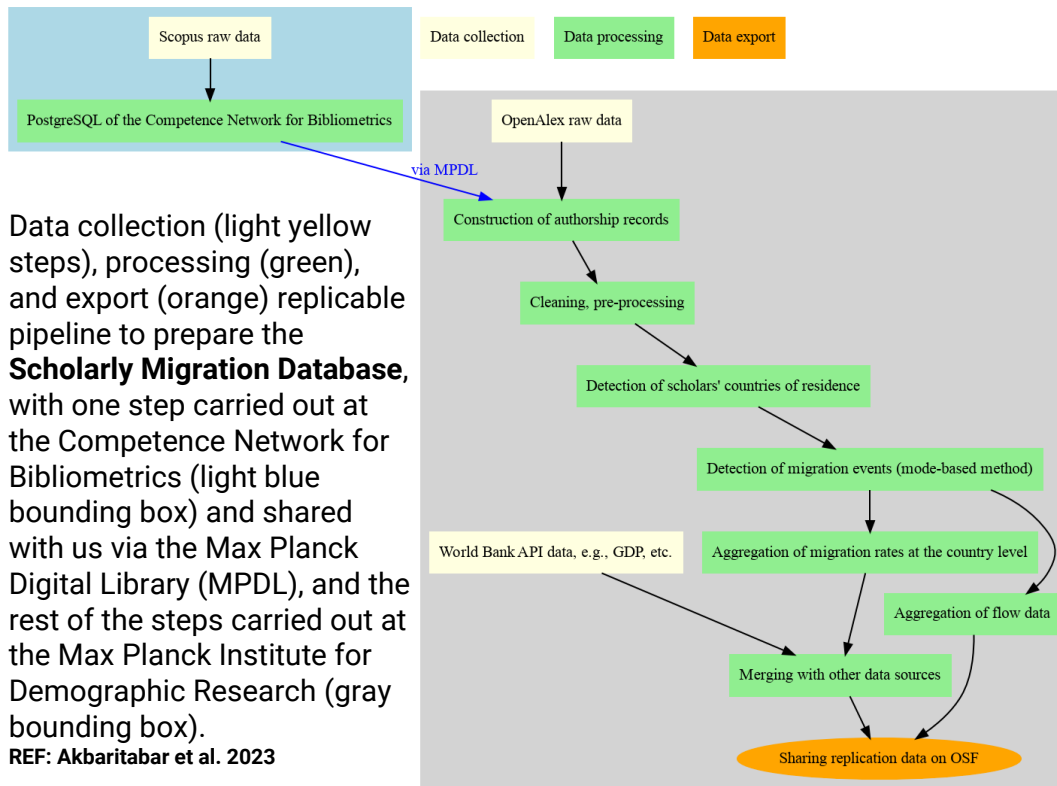
MAX PLANCK INSTITUTE
FOR DEMOGRAPHIC RESEARCH

Quantifying the Mobility and Migration of Scholars
Aliakbar Akbaritabar, the SMD core team, and collaborators

Why?

A lack of high-resolution and longitudinal data on migration of scholars worldwide and the unprecedented opportunity to repurpose scientific publications' metadata that spans more than 25 years.

How?



Data collection (light yellow steps), processing (green), and export (orange) replicable pipeline to prepare the **Scholarly Migration Database**, with one step carried out at the Competence Network for Bibliometrics (light blue bounding box) and shared with us via the Max Planck Digital Library (MPDL), and the rest of the steps carried out at the Max Planck Institute for Demographic Research (gray bounding box).

REF: Akbaritabar et al. 2023

What?

Scholarly Migration Database

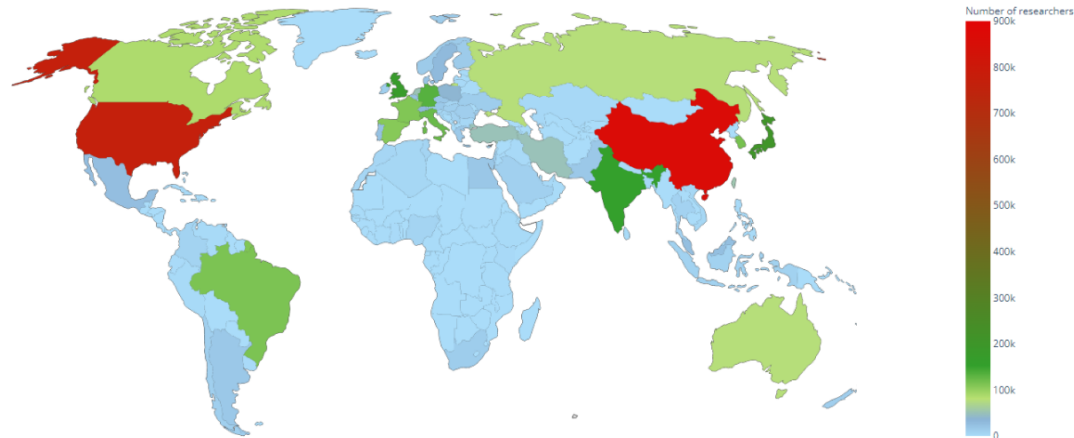
Quantifying the Mobility of Scholars

The Scholarly Migration Database provides detailed data about the mobility of scholars. Scientists, journalists and policy analysts who are interested in the movements of published researchers can access high-quality datasets and estimates via this website. The Scholarly Migration Database is prepared and hosted at the Max Planck Institute for Demographic Research in Rostock, Germany.

> [Research](#)

> [How is the data produced?](#)

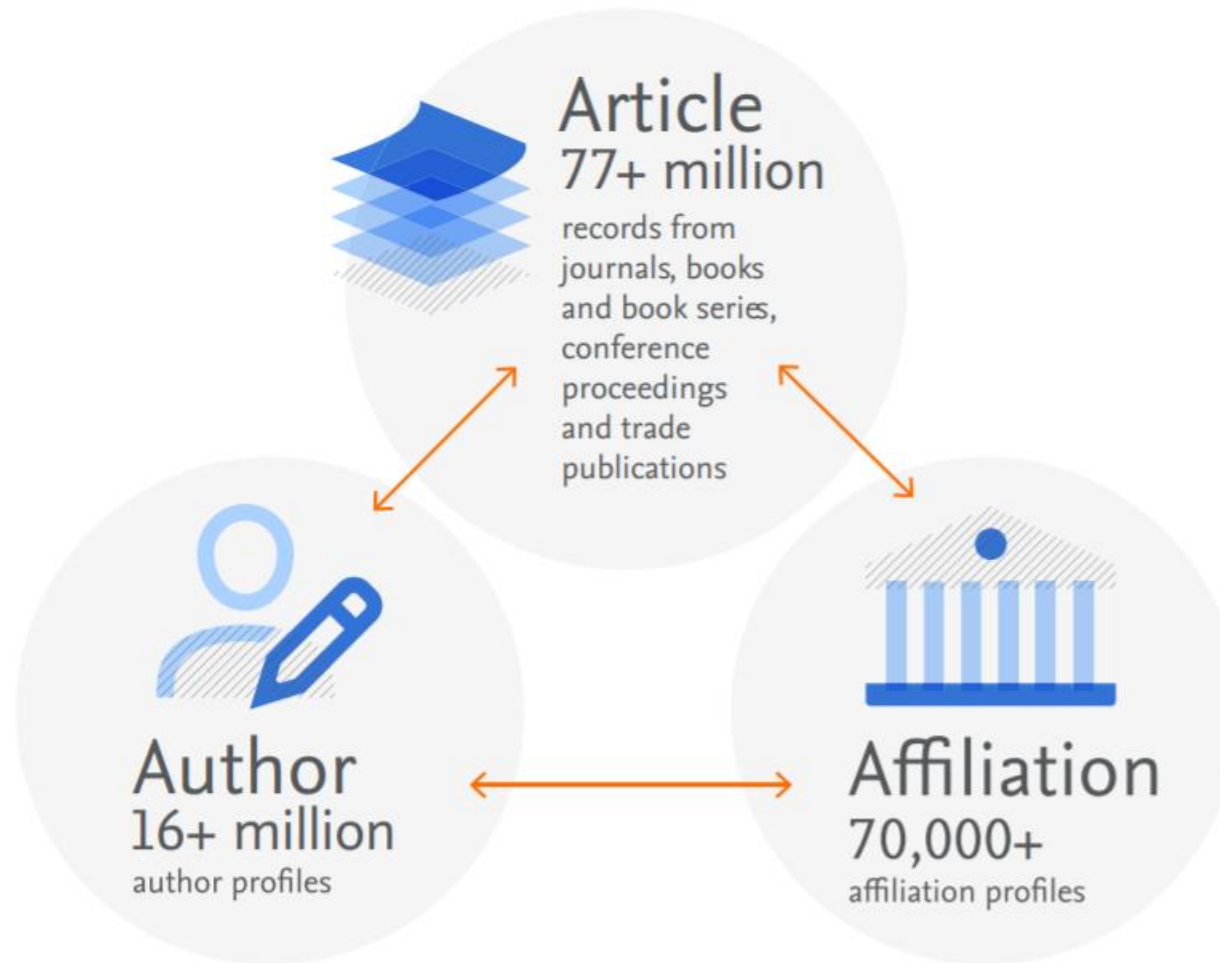
Number of published scholars per country
mean number of scopus published researchers in the years 2013 - 2017



A SIMPLE IDEA

Changes in institutional affiliations can be used to infer changes in residence over time for individual scholars and for populations

A SCALABLE IDEA: THE SCOPUS DATABASE



Data accessed via:



Source: https://www.elsevier.com/__data/assets/pdf_file/0007/69451/Scopus_ContentCoverage_Guide_WEB.pdf

Using 33 million Scopus publications
for all scientists worldwide

Using SMD data (so far): 6
publications, 6 working papers, and
multiple ongoing projects

Scholarly Migration Database
Quantifying the Mobility of Scholars

The Scholarly Migration Database provides detailed data about the mobility of scholars. Researchers and students who are interested in the movements of published researchers can download our datasets or explore the data on this website. The Scholarly Migration Database is hosted at the Max Planck Institute for Demographic Research in Rostock, Germany. You can [explore](#) and [download](#) the data here on this website.

[Download the data](#)
[Explore the data](#)
[How is the data produced?](#)

The 10 countries with the highest and lowest net migration, from 2000 to 2019, by gender and age. [View a map for details.](#)

Scholarly Migration Database
Quantifying the Mobility of Scholars

HOME EXPLORE RESEARCH DATA HISTORY TEAM

Core team (alphabetical order)
The Scholarly Migration Database is prepared, maintained and publicly released by the [Department of Digital and Computational Demography of the Max Planck Institute for Demographic Research \(MPDR\)](#).

Alakbar Akbaritabar
Alakbar Akbaritabar is a postdoctoral research scientist and the current deputy head for training at the Department of Digital and Computational Demography at the Max Planck Institute for Demographic Research. His research interests concern social networks, scholarly mobility, and networks of scientific collaboration.

Tom Theile
Tom Theile is a software developer at the Department of Digital and Computational Demography at the Max Planck Institute for Demographic Research.

Emilio Zagheni
Emilio Zagheni is Director of the Max Planck Institute for Demographic Research. He is best known for his research on combining digital trace data, traditional sources, and new forms of data collection, within solid statistical and formal demographic frameworks, in order to advance population science. In his various professional capacities, he has played a key role in fostering collaboration and exchange between demographers, statisticians and computer scientists.

Scholarly Migration Database
Quantifying the Mobility of Scholars

HOME EXPLORE RESEARCH DATA HISTORY TEAM

Selected publications and preprints using the Scholarly Migration Database and replication data

Sub-national disparities in the global mobility of academic talent
Alakbar Akbaritabar, Maciej J. Daňko, Xinyi Zhao and Emilio Zagheni, 2023, MPDR working paper <https://doi.org/10.4054/MPDR-WP-2023-038>

Global flows and rates of international migration of scholars
Alakbar Akbaritabar, Tom Theile and Emilio Zagheni, 2023, MPDR working paper <https://doi.org/10.4054/MPDR-WP-2023-018>
The code and data for replication are on [GitHub](#)

A gender perspective on the global migration of scholars
Xinyi Zhao, Alakbar Akbaritabar, Ridhi Kashyap and Emilio Zagheni, 2023, Proceedings of the National Academy of Sciences of the United States of America, 120 (10) <https://doi.org/10.1073/pnas.2214664120>
The code and data for replication are on [GitHub](#)

Global patterns of migration of scholars with economic development
Ebru Sanliturk, Emilio Zagheni, Maciej J. Daňko, Tom Theile, and Alakbar Akbaritabar, 2023, Proceedings of the National Academy of Sciences of the United States of America, 118 (25) <https://doi.org/10.1073/pnas.2217937118>
The code and data for replication are on [GitHub](#)

Scholarly Migration Database
Quantifying the Mobility of Scholars

Scholarly Migration Database Methods and Documentation
Version 2022_V1

Last updated: January 24, 2023 Version: 2022_V1

Contents

- 1 Summary 2
- 2 Scholarly Migration Database Team 2
- 3 Introduction 3
 - 3.1 Examples of research using Scholarly Migration Database 3
- 4 Bibliometric database 4
 - 4.1 Selection criteria for publications 4
 - 4.2 Scientific entity name disambiguation 4
 - 4.2.1 Author name 5
 - 4.2.2 Organization name 5
- 5 Pre-processing steps 5
 - 5.1 Construction of authorship records 5
 - 5.2 Clustering of origin assignments 6
 - 5.3 Gender assignment 7
- 6 Identifying migration events 7
 - 6.1 Multistep method 7
 - 6.2 Numerator and denominator populations 8
- 7 Measures 8
 - 7.1 How to use, cite, and contribute 8
 - 7.2 Data format 10
 - 7.3 How to download the data 10
 - 7.4 How to explore the data 10
 - 7.5 How to cite 10
 - 7.6 How to contribute 11

• Alakbar Akbaritabar, Max Planck Institute for Demographic Research (MPDR)
• Tom Theile, Max Planck Institute for Demographic Research (MPDR)
• Emilio Zagheni, Max Planck Institute for Demographic Research (MPDR)
January 24, 2023

Page 1

Scholarly Migration Database
Quantifying the Mobility of Scholars

HOME EXPLORE RESEARCH DATA HISTORY TEAM

Data exploration

Choose one or several countries
Click on the map or type:

Net migration numbers for: Brazil, China, Germany, France, United States

Net migration: In-migration Out-migration No researchers

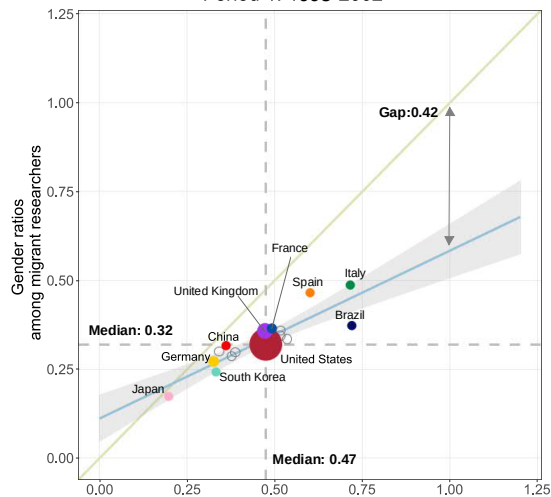
Absolute numbers Rates
Outflow Inflow

Scholarly Migration flow from Germany to Brazil, China, France, United States

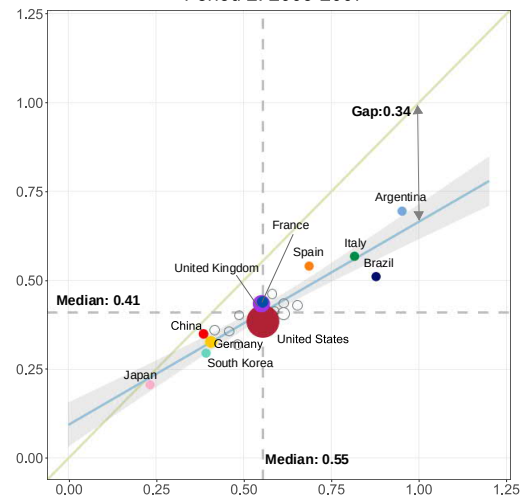
You can click on the map legend next to country names to remove it from the list or on the star to make it the focal country. Missing values mean that there were less than 5 migrations between these countries in a given year.



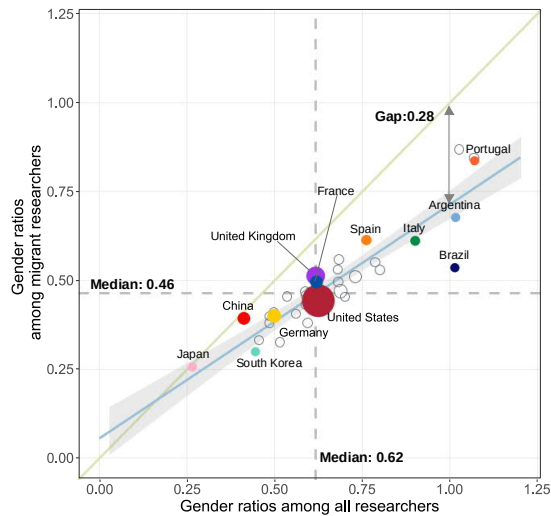
Period 1: 1998-2002



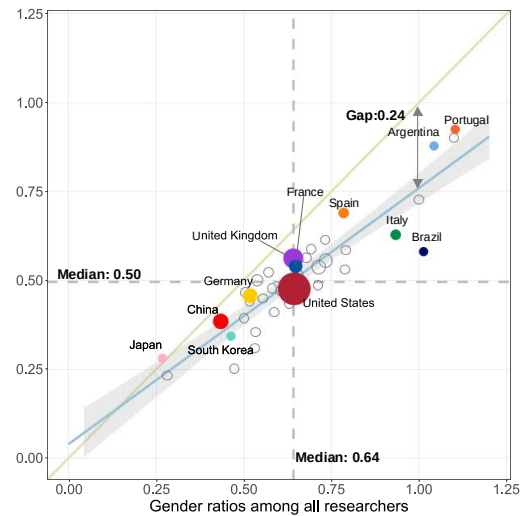
Period 2: 2003-2007

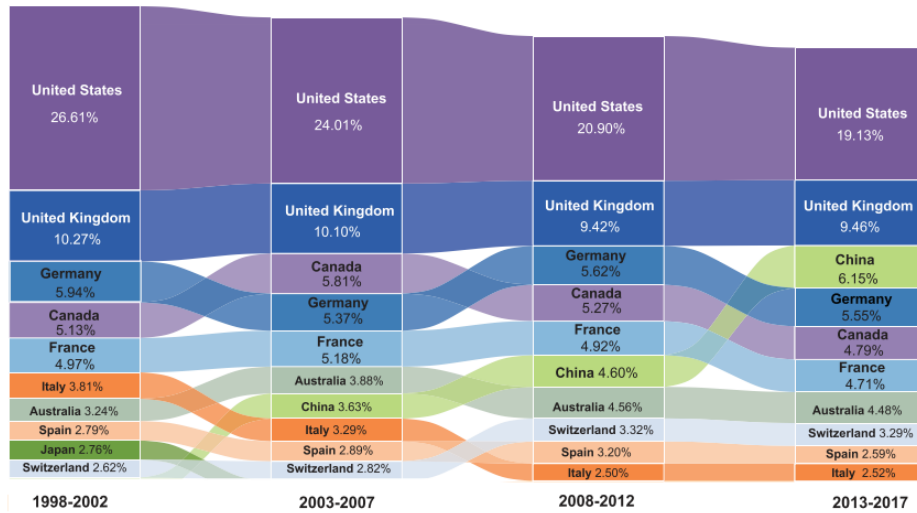


Period 3: 2008-2012

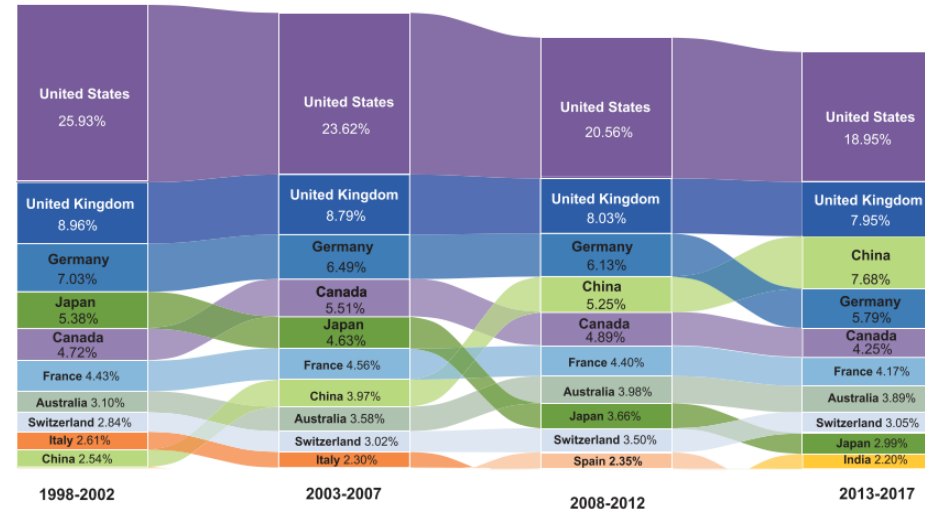


Period 4: 2013-2017





Female migrant researchers



Male migrant researchers

The 10 most preferred destination countries for female mobile researchers (left) and male mobile researchers (right).

REF: Zhao et al. 2023

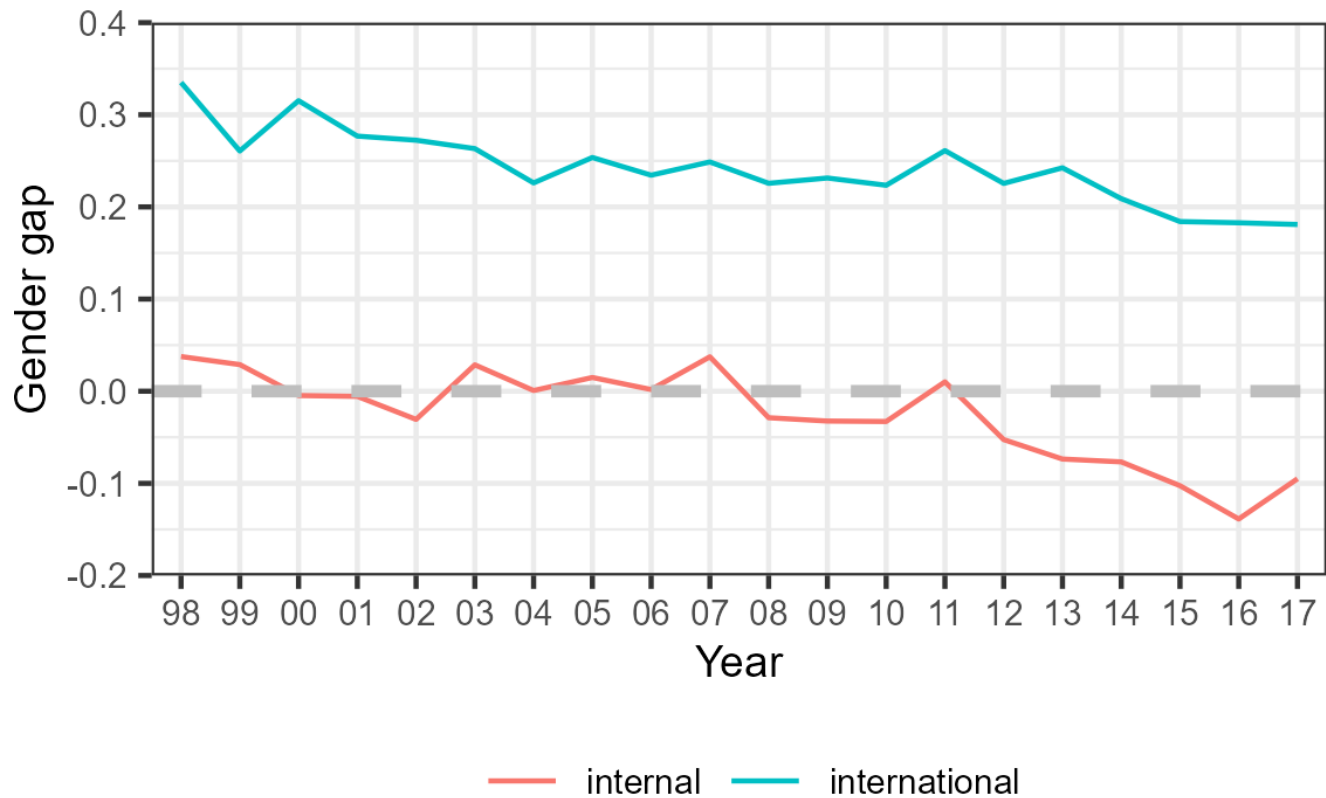
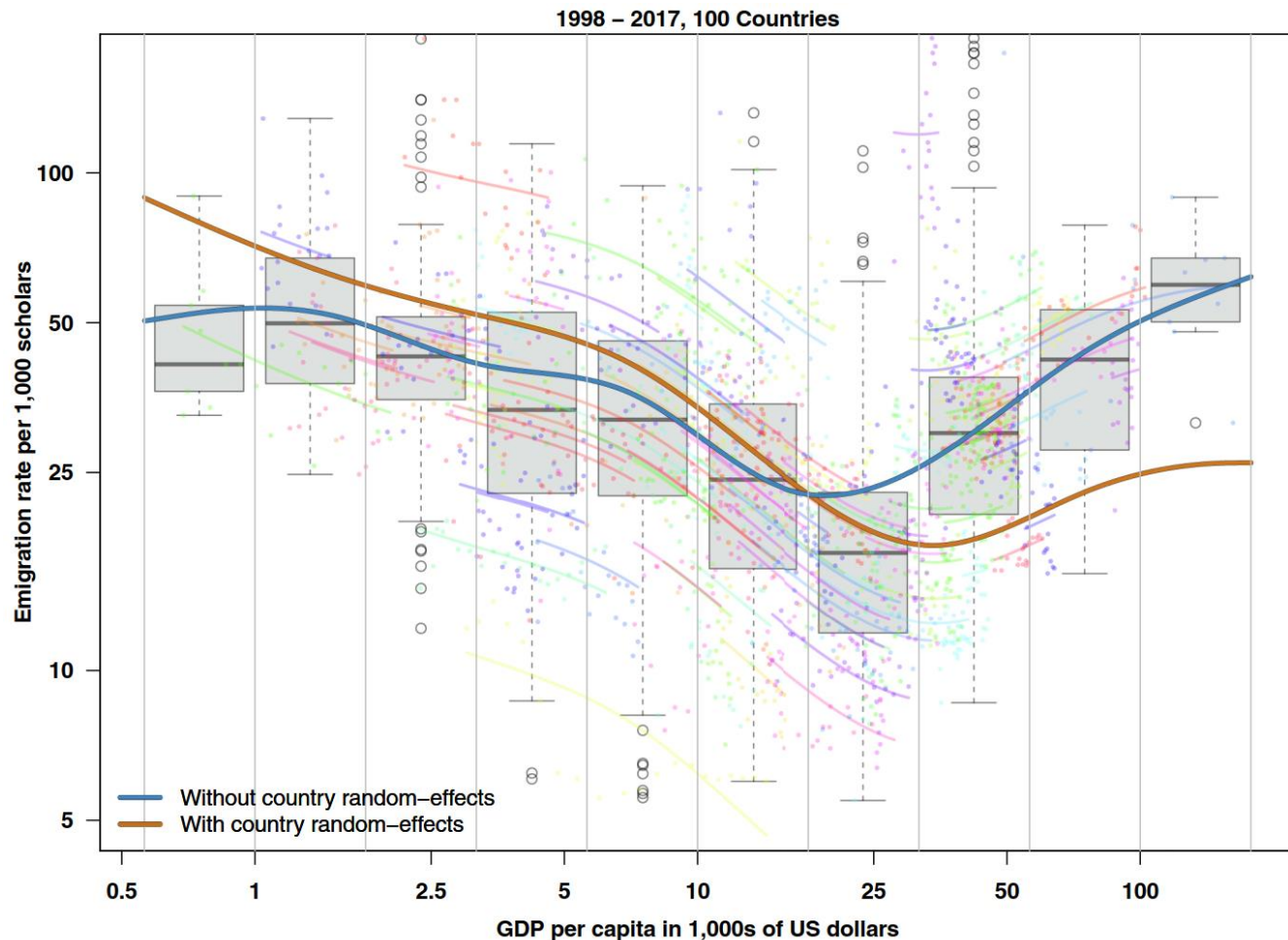
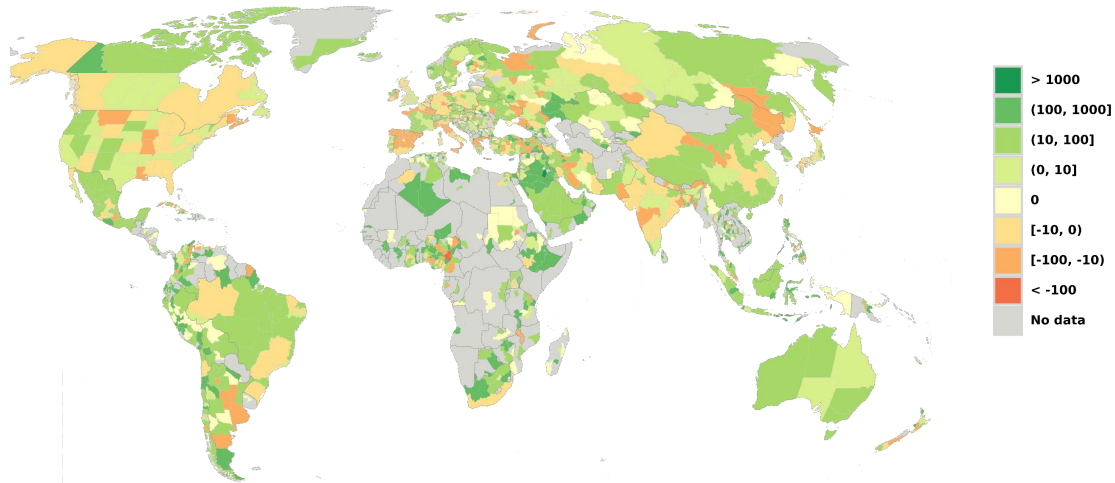


Fig. 3: Gender gap between internal (red) and international (blue) migrant scholars and all scholars by year. The dashed grey line indicates no differences between migrant scholars and all scholars

Scholarly emigration rates (1998 to 2017) as a function of GDP per capita in 2017 constant international dollars at purchasing power parity.

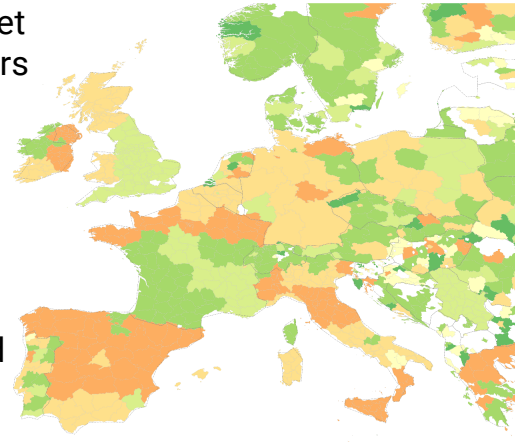
REF: Sanliturk et al. 2023



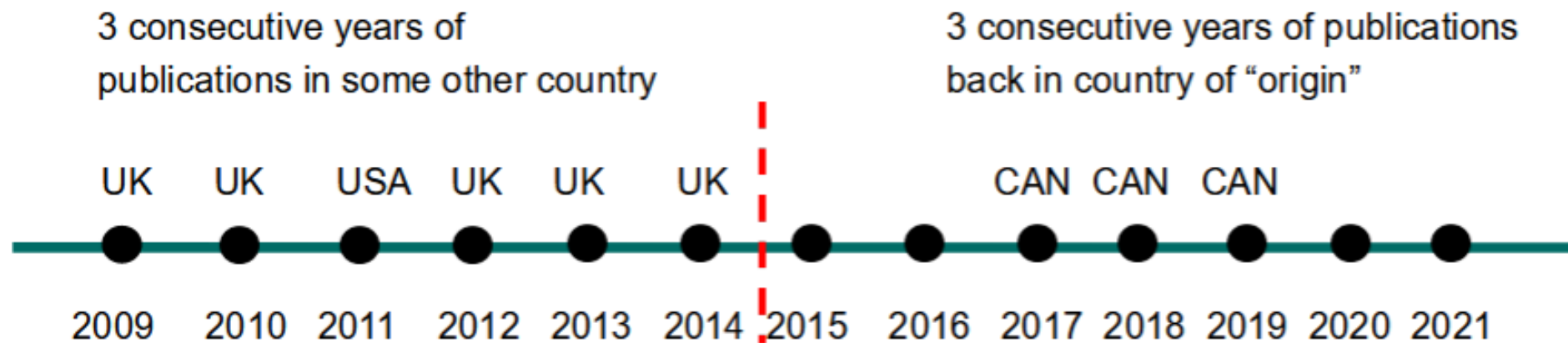
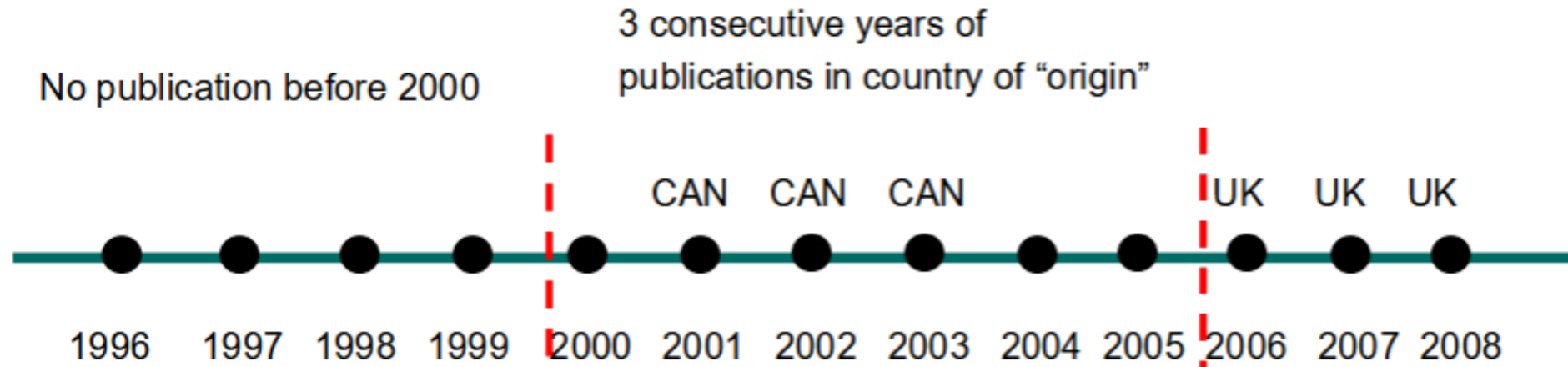


Combined internal and international net migration rate (NMR) per 1000 scholars at the sub-national level (GeoNames admin-1) worldwide (top) and Europe (right). Colors present the NMR of scholars sent (negative, red colors) or received (positive, green colors) in a year from 2012-2017 as an illustrative example of the 25 years analyzed, and yellow shows a balanced flow.

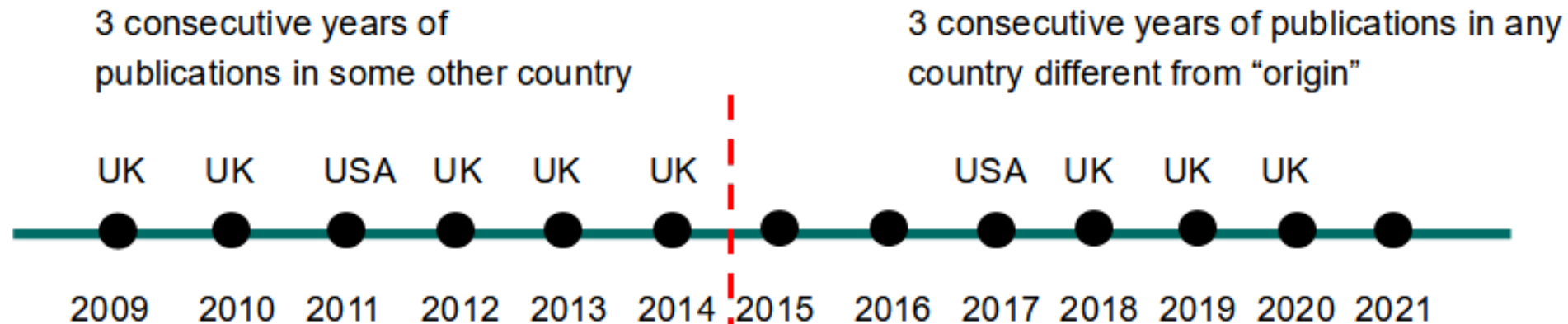
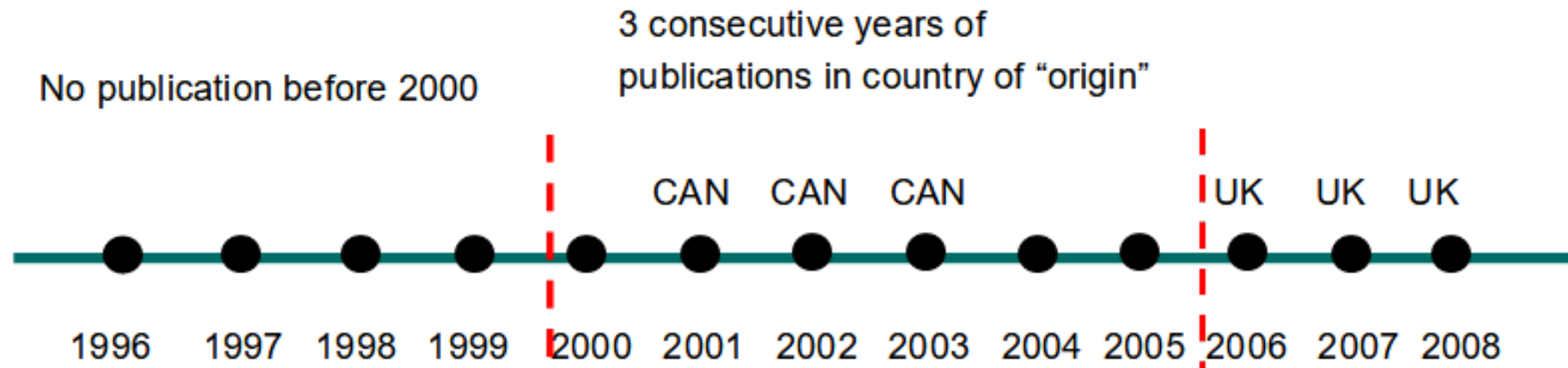
REF: Akbaritabar et al. 2023



Return Migration - Identifying return migration - Numerator



Return Migration - Identifying return migration - Denominator



Return Migration - Results

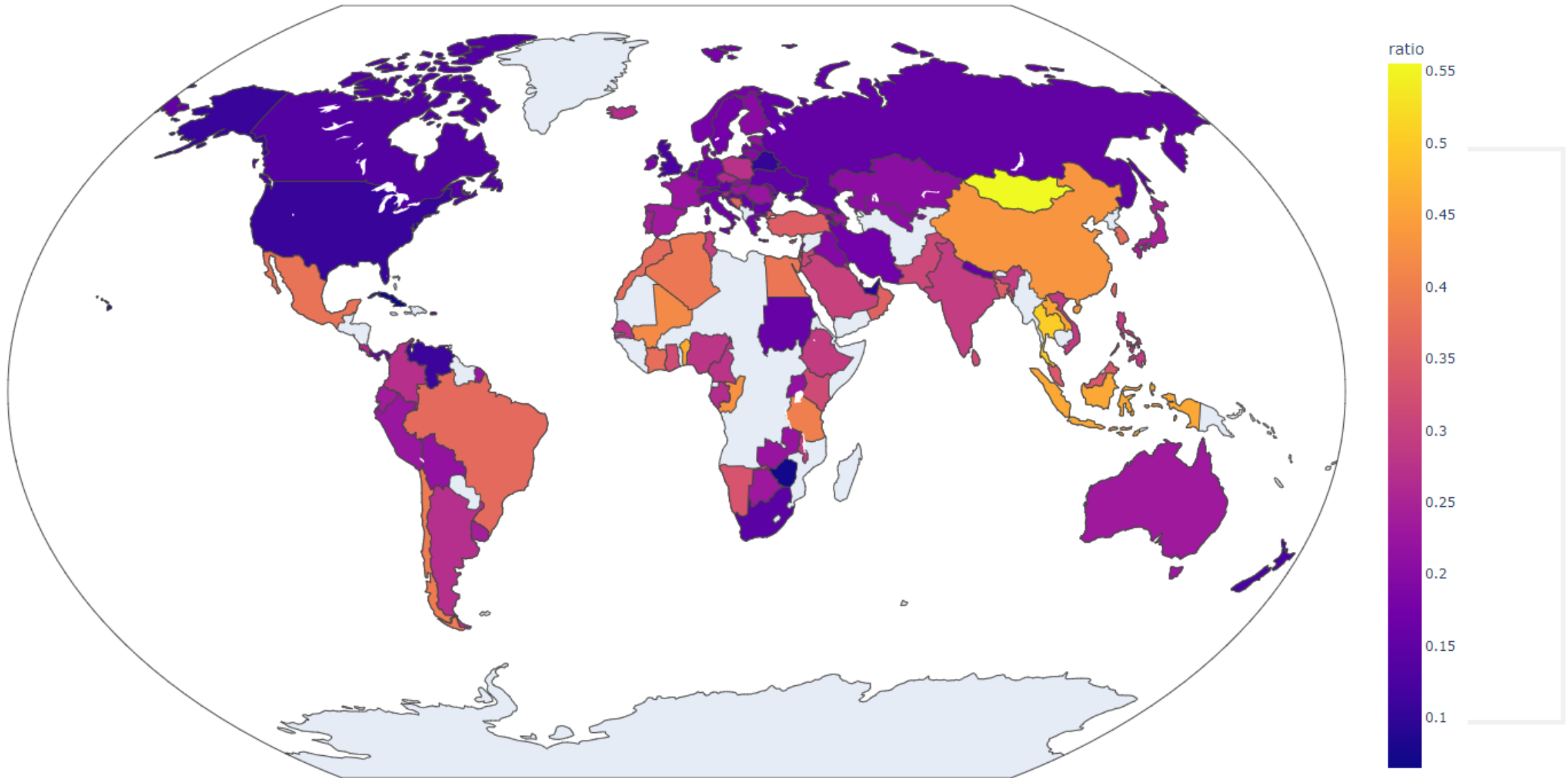




Fig. 1: Treemap of countries categorization in three levels from inner (lower granularity) to outer (higher granularity) circles.

Table 1: Wikipedia-based datasets used for training the Machine Learning model.

Name	Description	Volume
Wikipedia notable people’s mobility (29)	This dataset contains mobility data of notable people whose biography is registered in Wikipedia.	2,190,457
Nationality Estimate (11)	This dataset is constructed with name-nationality pairs by parsing the English-language Wikipedia.	696,136
WikiBio (Wikipedia Biography Dataset) (30)	This dataset gathers biographies from English Wikipedia.	418,978
Wikipedia Person Dataset (31)	This dataset gathers the infobox with the description of the person’s page.	355,468
Politicians on Wikipedia (32)	This dataset contains information about politicians from Wikipedia.	54,877
Pantheon 1.0 dataset (33)	This dataset includes the biographies present in more than 25 languages in Wikipedia.	16,071
Wikidata	Processed by our colleague Tom Theile.	131,731
	SUM	3,874,776
	SUM after removing duplicates	2,634,445

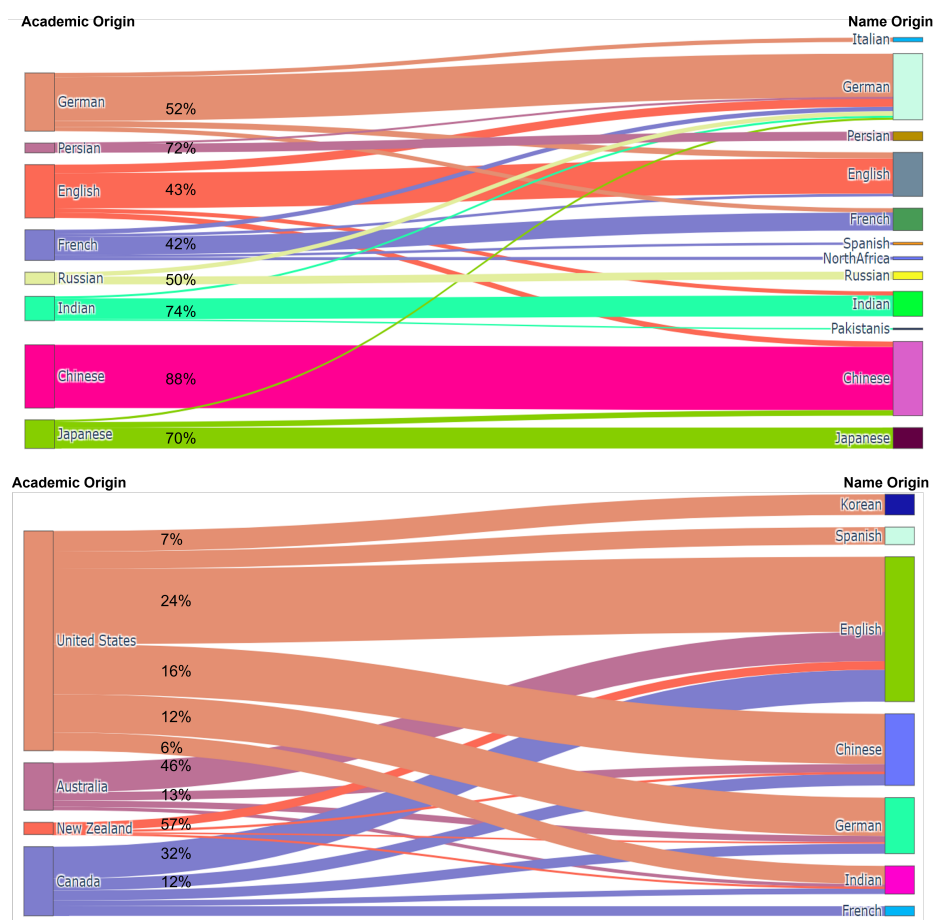


Fig. 3: Comparing the country of academic origin (left column) and assigned nationality using ML model (right column) at level 2 regions (top panel) and for the case of so-called immigration countries which have a diverse scientific workforce (bottom panel) that were excluded from training data.

Next?

- The next release of the SMD in 2024 will include disaggregated data by **gender, subnational regions** and **OpenAlex** equivalents to Scopus data
- A systematic calculation of return migration
- Considering international scholarly collaboration alongside mobility (i.e., if COVID inspired this as an alternative to geographical mobility?)

More?

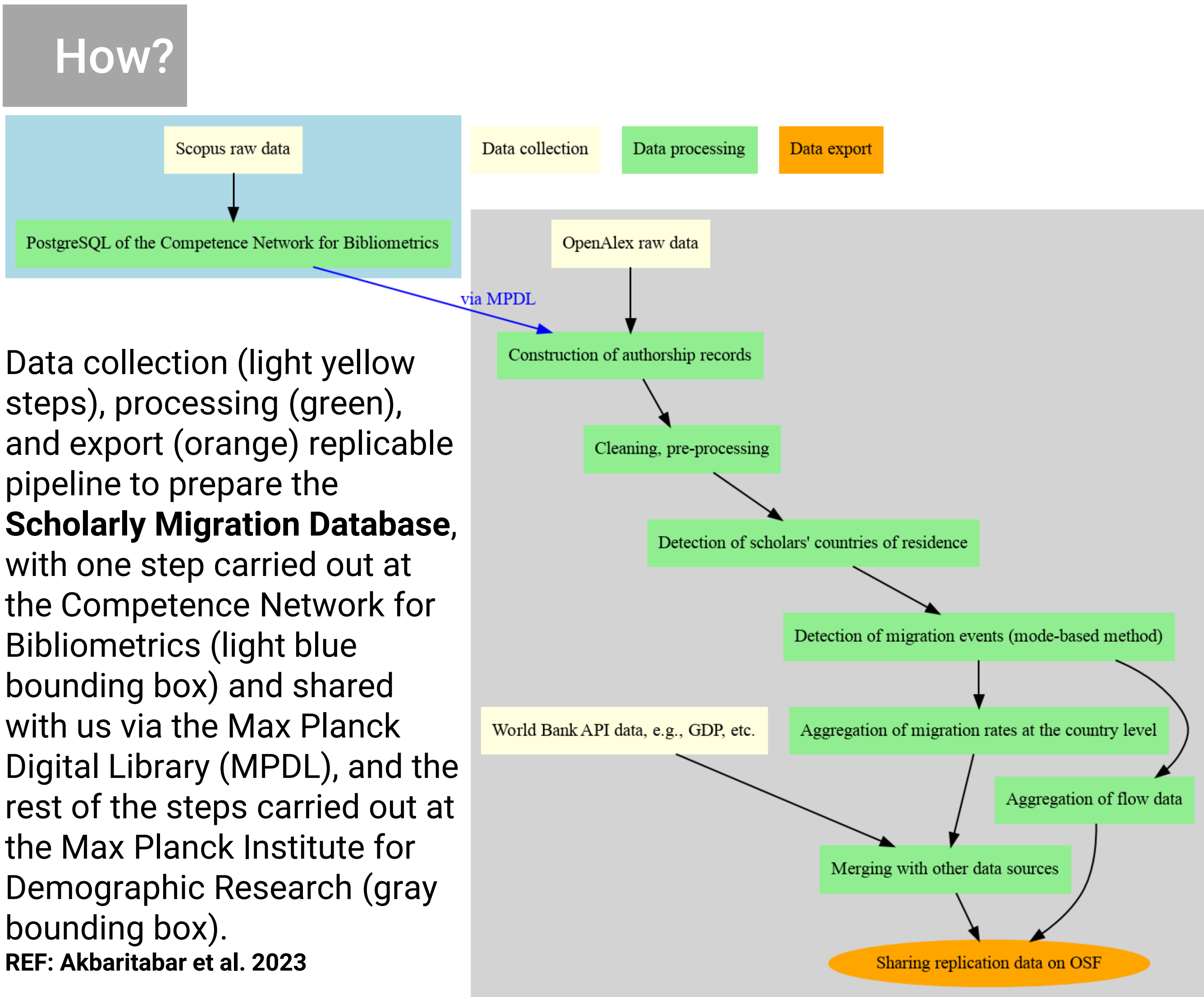


Scan me!

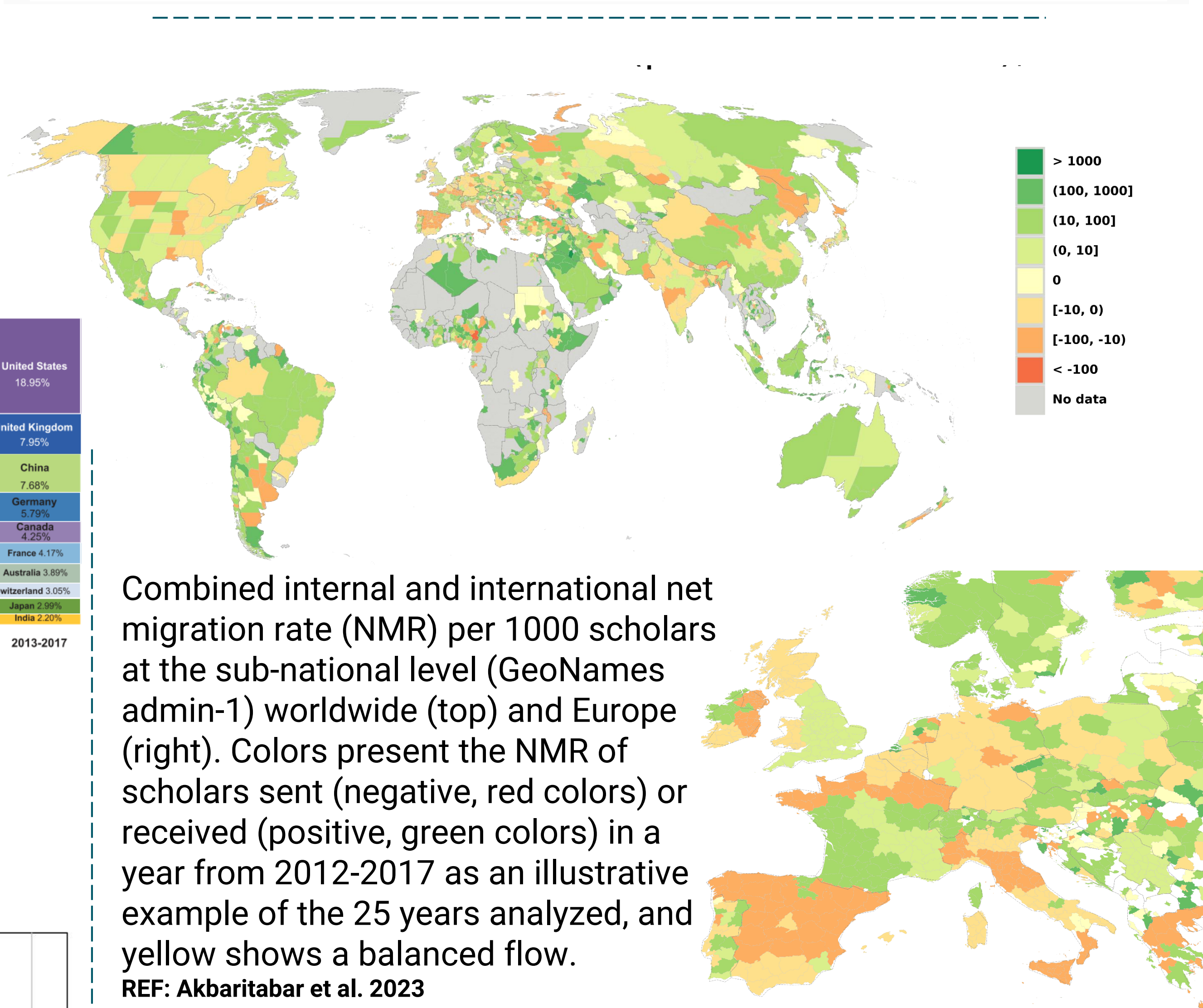
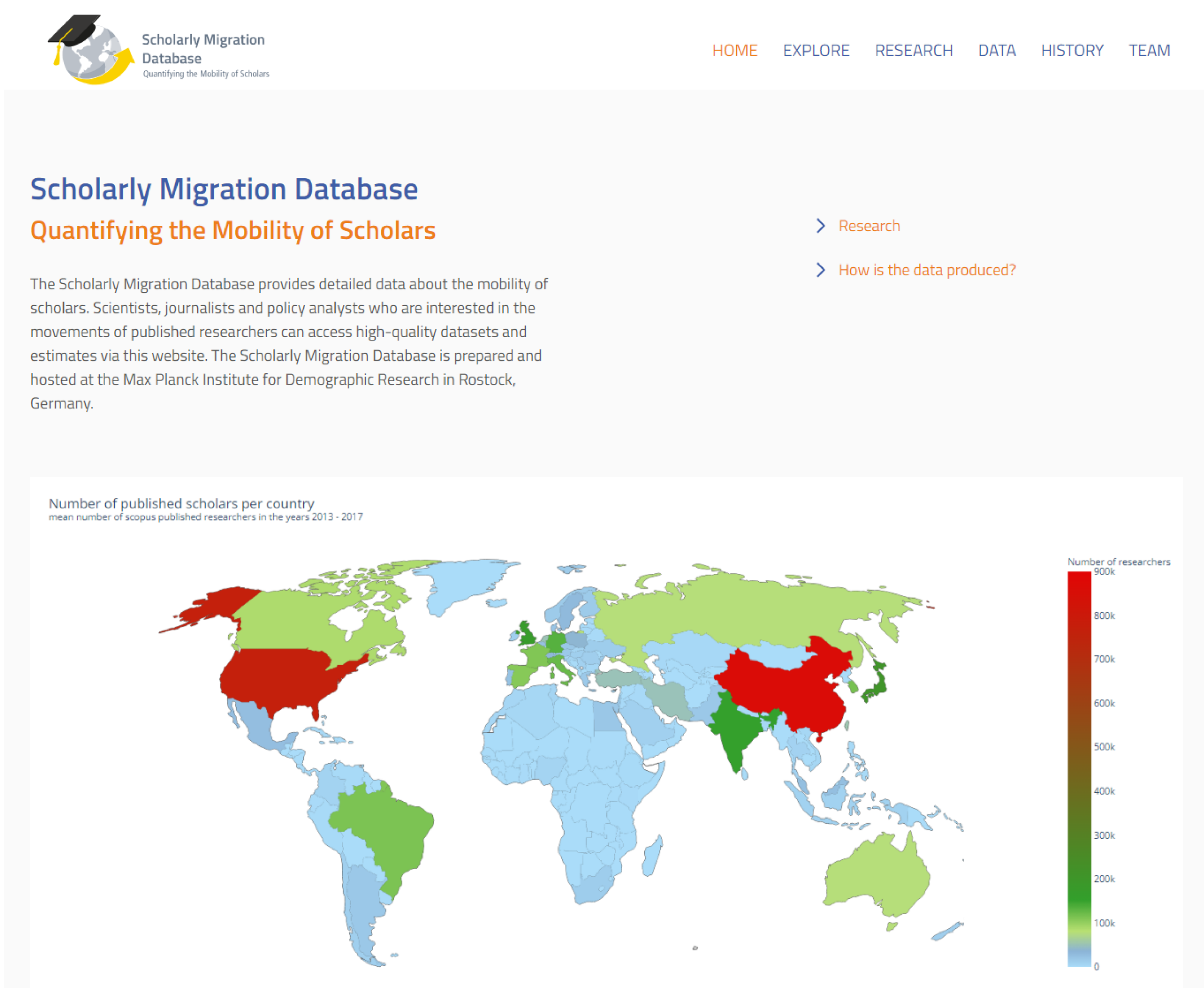


Quantifying the Mobility and Migration of Scholars
Aliakbar Akbaritabar, the SMD core team, and collaborators

Why? A lack of high-resolution and longitudinal data on migration of scholars worldwide and the unprecedented opportunity to repurpose scientific publications' metadata that spans more than 25 years.

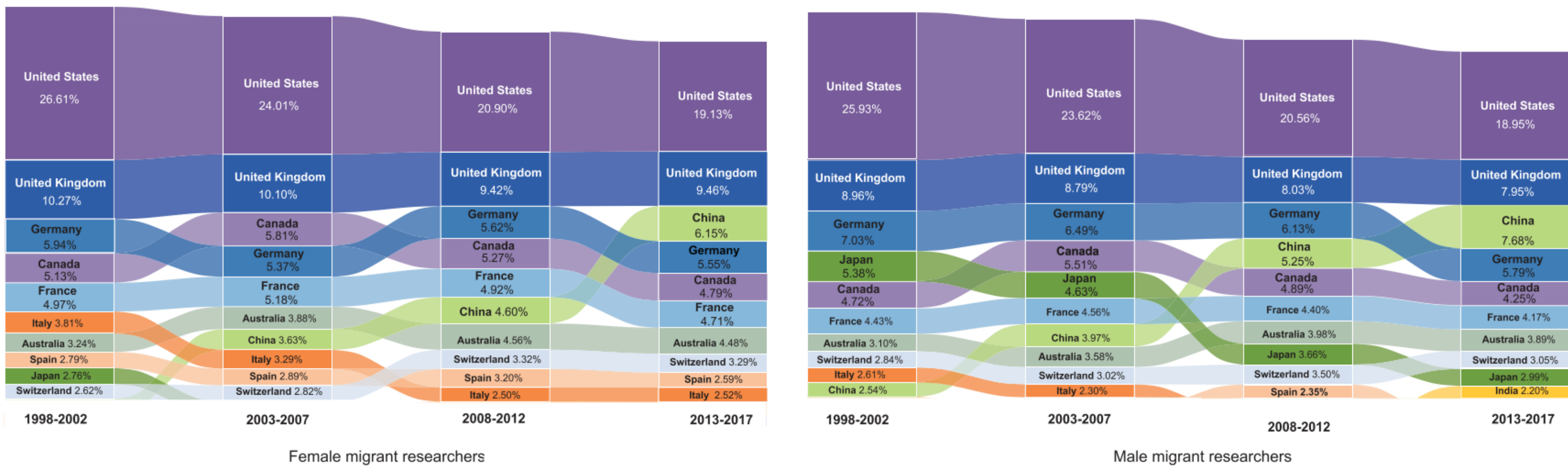


What?



Combined internal and international net migration rate (NMR) per 1000 scholars at the sub-national level (GeoNames admin-1) worldwide (top) and Europe (right). Colors present the NMR of scholars sent (negative, red colors) or received (positive, green colors) in a year from 2012-2017 as an illustrative example of the 25 years analyzed, and yellow shows a balanced flow.

REF: Akbaritabar et al. 2023

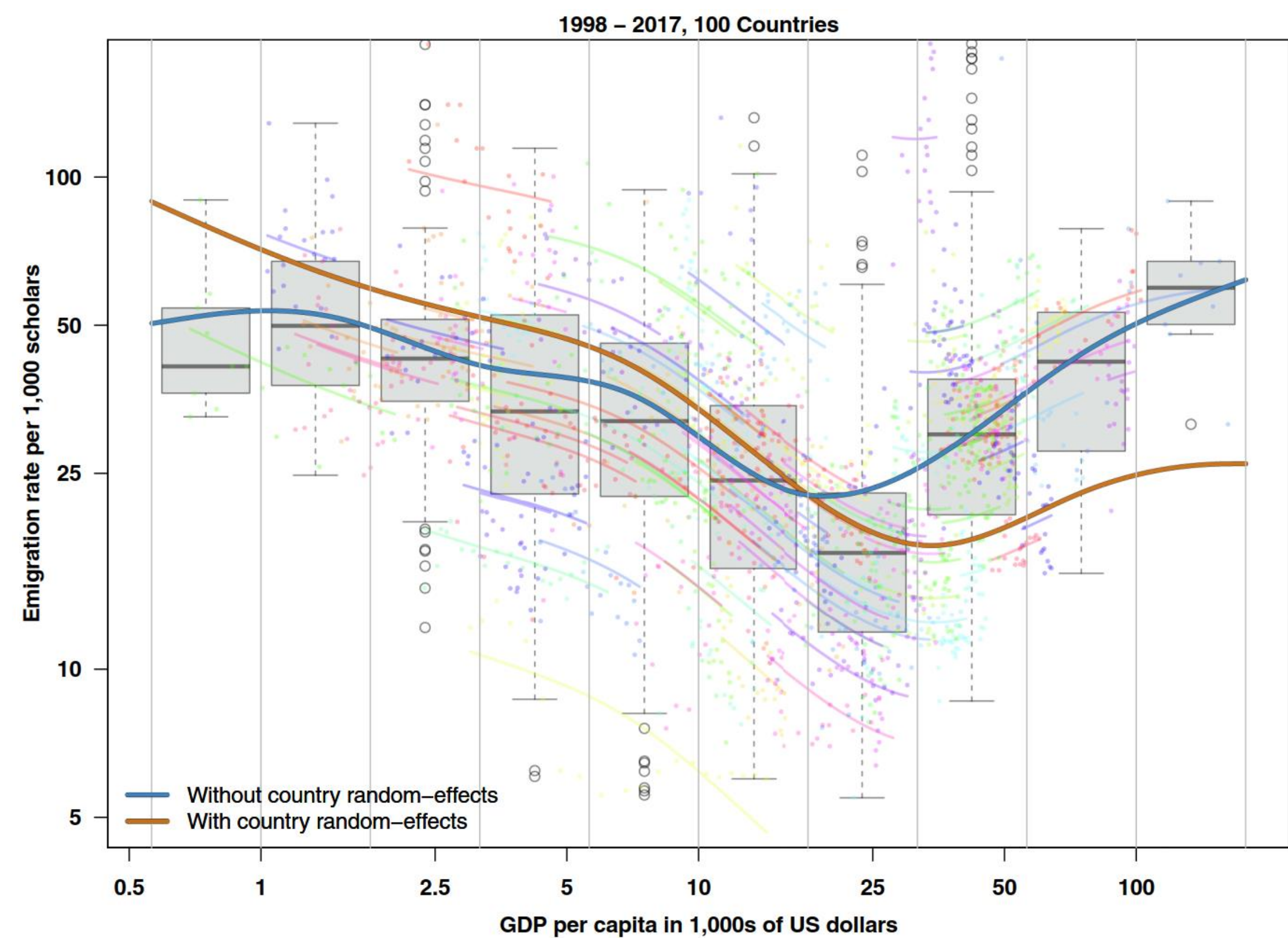


The 10 most preferred destination countries for female mobile researchers (left) and male mobile researchers (right).

REF: Zhao et al. 2023

Scholarly emigration rates (1998 to 2017) as a function of GDP per capita in 2017 constant international dollars at purchasing power parity.

REF: Sanliturk et al. 2023



- Next?**
- The next release of the SMD in 2024 will include disaggregated data by **gender**, **subnational regions** and **OpenAlex** equivalents to Scopus data
 - A systematic calculation of return migration
 - Considering international scholarly collaboration alongside mobility (i.e., if COVID inspired this as an alternative to geographical mobility?)

More?





Quantifying the Mobility and Migration of Scholars

Aliakbar Akbaritabar, the SMD core team, and collaborators

Why?

How?

A lack of high-resolution and longitudinal data on migration of scholars worldwide and the unprecedented opportunity to repurpose scientific publications' metadata that spans more than 25 years.

What?

Scholarly Migration Database

Quantifying the Mobility of Scholars

HOME

EXPLORE

RESEARCH

DATA

HISTORY

TEAM

> Research

> How is the data produced?

Scopus raw data

Data collection

Data processing

Data export

PostgreSQL of the Competence

Thank you for your attention!

Questions and comments are welcome!

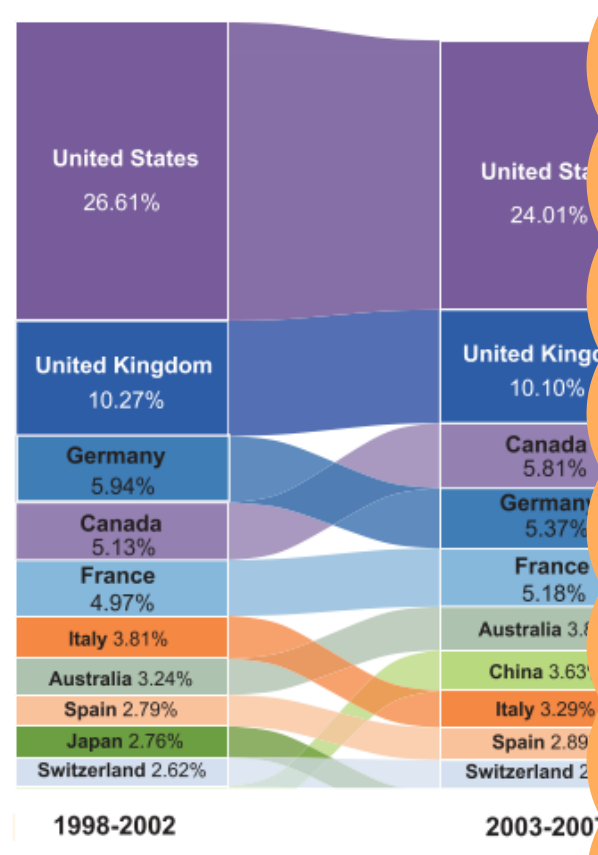
Aliakbar Akbaritabar

Research scientist and Deputy Head for Training

Max Planck Institute for Demographic Research (MPIDR), Rostock, Germany.

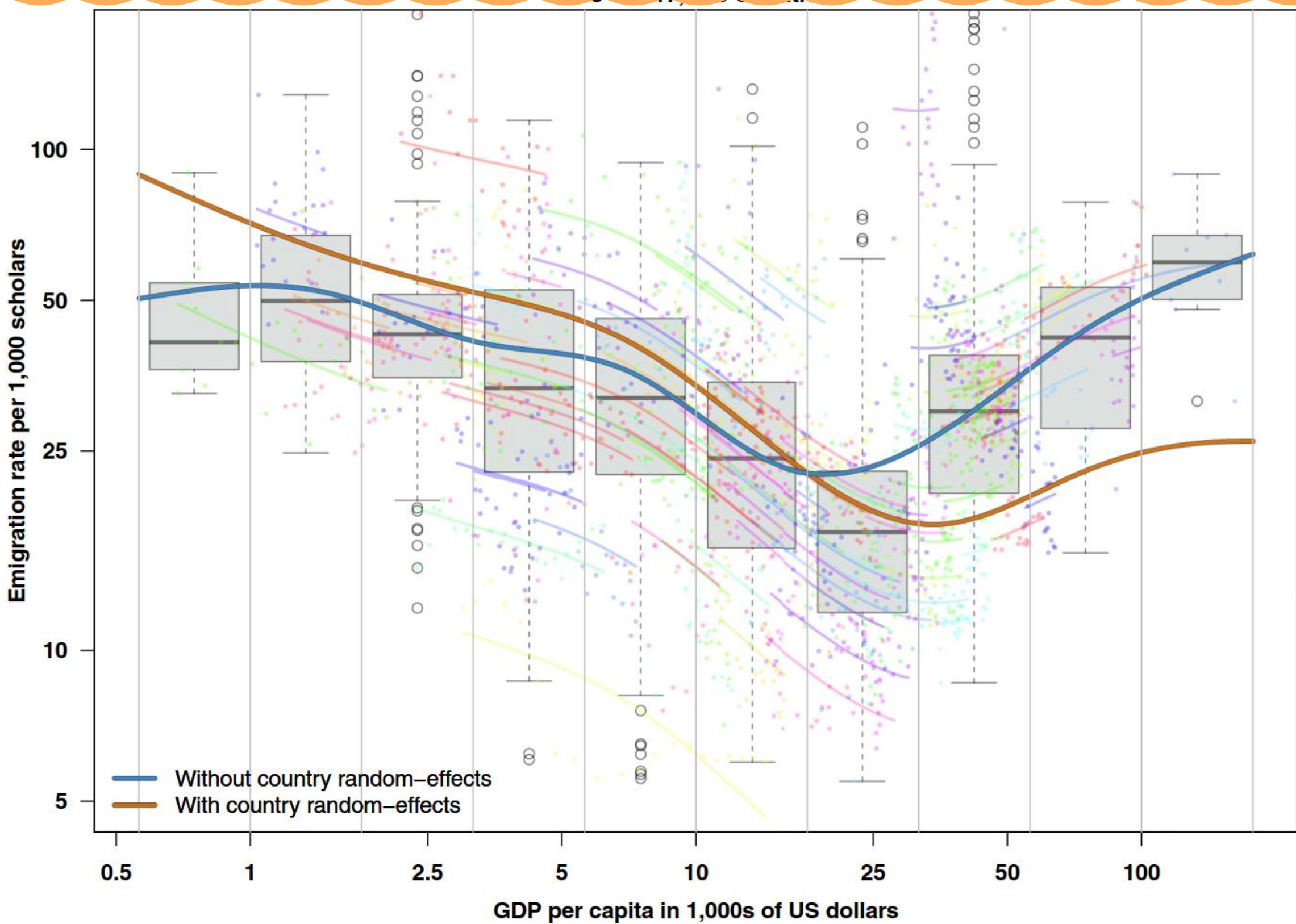
akbaritabar@demogr.mpg.de

Data collection steps), processing and export (or a pipeline to prepare the Scholarly Migration Database with one step call the Competence Bibliometrics (like a bounding box) and with us via the Max Planck Digital Library (MPDL). The rest of the steps are the Max Planck Demographic Research (MPDR) bounding box).
REF: Akbaritabar et al.



The 1st research
REF: Zha

Scholarly emigration rates (1998 to 2017) as a function of GDP per capita in 2017 constant international dollars at purchasing power parity.
REF: Sanliturk et al. 2023



yellow shows a balanced flow.
REF: Akbaritabar et al. 2023

Next?

- The next release of the SMD in 2024 will include disaggregated data by **gender**, **subnational regions** and **OpenAlex** equivalents to Scopus data
- A systematic calculation of return migration
- Considering international scholarly collaboration alongside mobility (i.e., if COVID inspired this as an alternative to geographical mobility?)



Study of scholarly migration at the MPIDR



- Akbaritabar, A., Dańko, M. J., Zhao, X., & Zagheni, E. (2023). Sub-national disparities in the global mobility of academic talent (WP-2023-038; 0 ed.). Max Planck Institute for Demographic Research. <https://doi.org/10.4054/MPIDR-WP-2023-038>
- Akbaritabar, A., Theile, T., & Zagheni, E. (2023). Global flows and rates of international migration of scholars (WP-2023-018; 0 ed., p. WP-2023-018). Max Planck Institute for Demographic Research. <https://doi.org/10.4054/MPIDR-WP-2023-018>
- Sanliturk, E., Zagheni, E., Dańko, M. J., Theile, T., & Akbaritabar, A. (2023). Global patterns of migration of scholars with economic development. *Proceedings of the National Academy of Sciences*, 120(4), e2217937120. <https://doi.org/10.1073/pnas.2217937120>
- Zhao, X., Akbaritabar, A., Kashyap, R., & Zagheni, E. (2023). A gender perspective on the global migration of scholars. *Proceedings of the National Academy of Sciences*, 120(10), e2214664120. <https://doi.org/10.1073/pnas.2214664120>
- Zhao, X., Aref, S., Zagheni, E., & Stecklov, G. (2022). Return migration of German-affiliated researchers: Analyzing departure and return by gender, cohort, and discipline using Scopus bibliometric data 1996-2020. *Scientometrics*
- Kashyap, R., Rinderknecht, R. G., Akbaritabar, A., Alburez-Gutierrez, D., Gil-Clavel, S., Grow, A., Kim, J., Leasure, D. R., Lohmann, S., Negraia, D. V., Perrotta, D., Rampazzo, F., Tsai, C.-J., Verhagen, M. D., Zagheni, E., & Zhao, X. (2022). *Digital and Computational Demography*. SocArXiv. <https://doi.org/10.31235/osf.io/7bvpt>
- Subbotin, A., & Aref, S. (2021). Brain drain and brain gain in Russia: Analyzing international migration of researchers by discipline using Scopus bibliometric data 1996–2020. *Scientometrics*. <https://doi.org/10.1007/s11192-021-04091-x>
- Miranda-González, A., Aref, S., Theile, T., & Zagheni, E. (2020). Scholarly migration within Mexico: Analyzing internal migration among researchers using Scopus longitudinal bibliometric data. *EPJ Data Science*, 9(1), 34. <https://doi.org/10.1140/epjds/s13688-020-00252-9>